

Upload CSV

Choose a CSV File

sample_data.csv

254.0B

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Rule Engine Specification

The agent dynamically scans any dataset:

1. Universal Checks:
- Duplicate rows.
 - Null or missing values in any column.
 - Mixed data types.
2. Text Formatting:
- Outer whitespace warnings.
3. Numeric Outliers:
- Mathematical outlier detection using the Interquartile Range (IQR).
 - Negative checks on positive count fields.
4. Standard Constraints:
- If `Age` is found: bounds checked [0, 100].
 - If `email` is found: regex format validated.
 - Missing checks in `name` and `roll number`.

API Secrets Configuration

Configure your OpenRouter API Key by creating:

- A `.streamlit/secrets.toml` file with `OPENROUTER_API_KEY="..."` OR
- A `.env` file in the root folder with `OPENROUTER_API_KEY="..."`

AI Data Quality Agent

Upload any CSV dataset to scan for data quality issues and anomalies. Our automated rule engine dynamically checks for missing values, duplicates, mixed types, email formats, and outliers, while **OpenRouter (Nex-AGI)** diagnoses patterns and writes custom Python cleanup scripts!

Dataset Preview

	name	rooll number	Age	ema
1	Bob Smith	None	22	bob
2	None	103	-5	chai
3	David Miller	104	35	davi
4	Eve Davis	105	28	Non
5	Frank Wilson	106	19	fran

Detected Column Attributes

Column Name	Inferred Type	Completeness (%)	Unique Values
name	Categorical	83.3%	
rooll number	Numeric (Integer)	83.3%	
Age	Numeric (Integer)	100.0%	
email	Email Address	83.3%	

Detected Errors

- ⚠ Detected 5 data quality warning(s)/anomaly(ies) in dataset:
- Row(s) 3: Missing name
 - Row(s) 2: Missing roll number
 - Row 3: Age is -5 (must be between 0 and 100)
 - Row(s) 5: Missing email
 - Row 2: Invalid email format 'bob.smith@invalid-email'

AI Suggestions

🔍 Diagnosis Summary

The anomalies look like data-entry and validation failures. Row 3 has a blank required field ('name') and an impossible Age value (-5), suggesting a missing required value plus a sign/transcription error. Row 2 has a missing 'rooll number' and an invalid email, while Row 5 has a blank email. The email issue for 'bob.smith@invalid-email' is specifically a missing top-level domain after the dot/TLD requirement. The missing roll number appears to be recoverable from the sequence 101, 103. 104. 105. so it is likely 102.